

Sunil Raut

Barrie, ON, Canada (Willing to Relocate) | rautsunil891@gmail.com | 249 989 4725 |

Hands on experience in machine learning with strong foundation in LLMs, NLP, and MLOps. Experienced in building and deploying end-to-end ML and LLM-powered applications using Python, FastAPI, Hugging Face, and PyTorch. Passionate about operationalizing AI systems from data preparation to scalable deployment and ensuring model reliability, observability, and continuous improvement.

AREAS OF EXPERTIS

Programming: Python, SQL, FastAPI, Flask, Bash, ML / AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLM / NLP Techniques: Prompt Engineering (Zero/Few-Shot), RAG, Fine-Tuning, CoT, ReAct, Data & Tools: Pandas, NumPy, Airflow, Kafka, Power BI, Tableau, Databases: MongoDB, MySQL, PostgreSQL, Pinecone (Vector DB), Cloud / DevOps: Azure, AWS, Docker, Git, CI/CD, Supabase ,Other Skills: Model Evaluation, Data Pipelines, API Development, Model Monitoring.

PROFESSIONAL EXPERIENCE

Data Engineer

Nepdroid Innovations Pvt. Ltd., Nepal

June 2022– April 2024

- Designed and implemented **Azure Data Factory** and **Databricks** pipelines for enterprise analytics.
- Built automated ETL workflows to integrate APIs, SQL databases, and cloud storage systems.
- Contributed to **metadata management and data lineage tracking** using Microsoft Purview.
- Supported ML team by preparing datasets, maintaining model pipelines, and documenting data workflows

EDUCATION

- POST GRADUATE DIPLOMA- BIG DATA ANALYTICS AND ARTIFICIAL INTELLIGENCE DESIGN & IMPLEMENTATION** Sep 2023- Apr 2025
Georgian College, Barrie, Ontario, Canada
- BACHELOR'S DEGREE- ELECTRONICS AND COMMUNICATION ENGINEERING** Nov 2016- Apr 2021
Institute of Engineering, Tribhuvan University, Nepal

PROJECTS AND LEADERSHIP

EVA –Academic Content Assistant (LLM-Powered AI System) – Tarsi Group, Barrie, ON, Canada

Jan 2025-Apr 2025

Tech Stack: FastAPI, Groq (Llama 3.2-90B Vision Preview), Supabase, Python, Docker

- Developed **EVA**, an intelligent **academic content assistant** that generates course outlines, lesson plans, assignments, and rubrics through **natural conversations**.
- Integrated **Groq's LLM (Llama 3.2-90B)** with **FastAPI** to power conversational academic content generation.
- Implemented **secure user authentication** using **Supabase**, and conversation history management for personalized sessions.
- Enabled **automatic Word document generation** for downloadable academic outputs.
- Built a **modern, responsive UI** for seamless interaction with the backend API.
- Optimized prompt engineering and response consistency for domain-specific academic tasks.

Impact: Demonstrated full-stack LLM application design from prompt optimization and inference handling to secure API delivery and scalable deployment.

Stock Price Prediction Web Application

Tech Stack: Python, FastAPI, LSTM (Keras/TensorFlow), Scikit-learn, Pandas

- Built a **FastAPI-based web application** to predict stock prices using an **LSTM model** trained on historical stock data.
- Integrated automatic **data preprocessing and scaling** with a pre-fitted Scikit-learn scaler.
- Designed a **modular prediction API** for real-time inference, enabling seamless integration into dashboards.
- Implemented REST endpoints for stock symbol selection, model inference, and performance monitoring.

IMDB Movie Review Sentiment Analysis

Tech Stack: Scikit-learn, TF-IDF, XGBoost, Random Forest, GridSearchCV

- Developed a binary sentiment classifier on **25K IMDB reviews**, performing **text preprocessing** and **TF-IDF vectorization**.
- Optimized model performance via **GridSearchCV**, improving overall accuracy and interpretability.

CERTIFICATIONS

- Python – Udemy
- MS SQL – LinkedIn Learning
- Machine Learning Fundamentals - Data camp
- Apache Spark by example-LinkedIn Learning
- AWS Academy Machine Learning Foundation
- Databricks Core Concept -Databricks

SKILLS

Analytical Thinking, Adaptability, Accountability, Problem Management, Change Management, Time Management, Task Prioritization, Communication, Organized, Documentation, Attention to detail.